



## **Part A -Solution**

### **❑ What is chemical bond?**

#### **Answer:**

**Chemical bonding** is the attractive force between atoms that holds them together in molecules and compounds. This force arises from the interaction of electrons between atoms.

There are three main types of chemical bonding: ionic, covalent, and metallic.

### **❑ What is ionic bond?**

#### **Answer:**

Ionic bonding is a type of chemical bond that occurs between a metal and a nonmetal. In ionic bonding, one atom donates electrons to another atom, resulting in the formation of ions with opposite charges. The electrostatic attraction between these ions holds the compound together.

#### **Examples of ionic compounds:**

- Sodium chloride (NaCl)
- Potassium bromide (KBr)
- Calcium oxide (CaO)
- Magnesium sulfate (MgSO<sub>4</sub>)

### ❑ What is covalent bond?

#### Answer:

Covalent bonding is a type of chemical bond where two or more atoms share electrons to attain a stable electron configuration. This sharing of electrons results in a strong attractive force between the atoms, holding them together in a molecule.

#### **Examples of covalent compounds:**

- Water ( $\text{H}_2\text{O}$ )
- Carbon dioxide ( $\text{CO}_2$ )
- Methane ( $\text{CH}_4$ )
- Ammonia ( $\text{NH}_3$ )
- Glucose ( $\text{C}_6\text{H}_{12}\text{O}_6$ )

### ❑ What is co-ordinate covalent bond?

#### Answer:

A coordinate covalent bond, also known as a dative covalent bond, is a type of covalent bond where one atom donates both electrons to form a shared pair with another atom. In this type of bond, one atom acts as the electron donor and the other acts as the electron acceptor.

### Examples:

- Ammonium ion ( $\text{NH}_4^+$ ): The nitrogen atom donates a lone pair of electrons to form a coordinate covalent bond with a hydrogen ion.
- Carbon monoxide (CO): The carbon atom donates a lone pair of electrons to form a coordinate covalent bond with the oxygen atom.

**❑ What is orbital hybridization? What is sp/sp<sup>2</sup>/sp<sup>3</sup> hybridization?**

### Answer:

**Orbital hybridization** is a quantum mechanical theory that explains the bonding of atoms in molecules. It involves the mixing of atomic orbitals to form new hybrid orbitals that are more suitable for bonding.

#### **❑ sp Hybridization:**

- One's orbital and one p orbital combine to form two sp hybrid orbitals.
- These hybrid orbitals are arranged at 180° angles.
- Examples:  $\text{BeCl}_2$ ,  $\text{CO}_2$

#### **❑ sp<sup>2</sup> Hybridization:**

- One's orbital and two p orbitals combine to form three  $sp^2$  hybrid orbitals.
- These hybrid orbitals are arranged at  $120^\circ$  angles.
- Examples:  $CH_4$ ,  $C_2H_4$

#### □ $sp^3$ Hybridization:

- One's orbital and three p orbitals combine to form four  $sp^3$  hybrid orbitals.
- These hybrid orbitals are arranged at  $109.5^\circ$  angles.
- Examples:  $CH_4$ ,  $NH_3$ ,  $H_2O$

#### □ What is (VSEPR) theory?

##### Answer:

The **Valence Shell Electron Pair Repulsion (VSEPR) theory** is a model used to predict the molecular geometry of molecules based on the repulsion between electron pairs (bonding and lone pairs) around a central atom. The theory assumes that electron pairs will arrange themselves in a way that minimizes the repulsion between them.

## Key principles of VSEPR theory:

- **Electron pairs repel each other:** Bonding pairs and lone pairs of electrons around a central atom will arrange themselves in a way that minimizes the repulsion between them.
- **Bonding pairs and lone pairs are equivalent:** Both bonding pairs and lone pairs are treated as electron clouds that repel each other.
- **Lone pairs are more repulsive than bonding pairs:** Lone pairs of electrons are generally more repulsive than bonding pairs.

## ❑ What are Transition Elements?

### Answer:

Transition elements (also known as transition metals) are elements that have partially filled d orbitals. IUPAC defines transition elements as an element having a d subshell that is partially filled with electrons, or an element that has the ability to form stable cations with an incompletely filled d orbital.

In general, any element which corresponds to the d-block of the modern periodic table (**which consists of groups 3-12**) is considered to be a transition element.

❑ **All transition elements are d- block elements, but all d - block elements are not transition elements.**

**Answer:**

Elements in which the last electron enters in the d orbitals are called d - block elements or transition elements, these elements have the general outer electronic configuration  **$(n-1)d(1-10)ns(0-2)$**  Zn, Cd and Hg having the electronic configuration  **$(n-1)d^{10}ns^2$**  Do not show most of the properties of transition elements.

The d- orbitals in these elements are completely filled in the ground state as well as in their common oxidation states. Therefore, they are not regarded as transition elements.

Thus, on the basis of properties, all transition elements are d - block elements but on the basis of electronic configuration, all d - block elements are not called transition elements.

**Part-B Solution**

❑ **Discuss ionic/covalent/Coordinate bond with Suitable example.**

**Answer:**

### **Ionic Bonding**

- **Occurs between:** A metal and a nonmetal.
- **Process:** The metal atom loses electrons to become a cation, while the nonmetal atom gains electrons to become an anion. The oppositely charged ions are attracted to each other, forming an ionic bond.
- **Example:** Sodium chloride (NaCl)
- **Characteristics:** High melting points, hard solids, soluble in water, and conduct electricity in solution.

### **Covalent Bonding**

- **Occurs between:** Two nonmetals.
- **Process:** Atoms share electrons to achieve a stable electron configuration.
- **Example:** Water (H<sub>2</sub>O)
- **Characteristics:** Lower melting points compared to ionic compounds, can be solids, liquids, or gases, and generally do not conduct electricity.

## Coordinate Covalent Bonding

- **Occurs between:** One atom donates both electrons to form a shared pair with another atom.
- **Example:** Ammonium ion ( $\text{NH}_4^+$ )
- **Characteristics:** Unequal sharing of electrons, often found in complex ions or molecules.

**Add some description from part A same question.**

- ☐ **Write the properties of transition element and example two of them.**

**Answer:**

Transition elements (also known as transition metals) are elements that have partially filled d orbitals. IUPAC defines transition elements as an element having a d subshell that is partially filled with electrons, or an element that has the ability to form stable cations with an incompletely filled d orbital.

In general, any element which corresponds to the d-block of the modern periodic table (**which consists of groups 3-12**) is considered to be a transition element.

## Properties of Transition Elements

Transition elements are a group of chemical elements that occupy the d-block of the periodic table. They are located between the s-



block elements and the p-block elements. Here are some key properties of transition elements:

- **Variable oxidation states:** Transition elements can exhibit variable oxidation states, meaning they can form ions with different charges. This is due to the relatively close energy levels of their 4s and 3d orbitals.
- **Colored compounds:** Many transition metal compounds are colored due to the d-d transitions of electrons.
- **Catalytic properties:** Transition metals often have catalytic properties, meaning they can increase the rate of chemical reactions without being consumed.
- **Para magnetism:** Many transition elements are paramagnetic, meaning they are attracted to a magnetic field.
- **Formation of complexes:** Transition metals can form complex ions with ligands, which are molecules or ions that bond to the metal ion.
- **High melting and boiling points:** Transition metals generally have high melting and boiling points due to the strong metallic bonding between their atoms.
- **Hard and ductile:** Many transition metals are hard and ductile, meaning they can be shaped without breaking.

### Examples of Transition Elements:

1. **Iron (Fe):** Iron is a common transition metal with variable oxidation states. It is a key component of steel and is essential for human health.

2. **Copper (Cu):** Copper is another important transition metal with a wide range of applications, including electrical wiring, plumbing, and coins. It is also an essential nutrient for humans.

These are just a few examples of transition elements, and there are many others with unique properties and applications.

- ❑ **Discuss hybridization  $\text{BeCl}_2/\text{BeCl}_3$  /  $\text{CH}_4$ . it's structure and bond angle.**

**Answer:**

**Hybridization** is a quantum mechanical concept that explains the bonding of atoms in molecules. It involves the mixing of atomic orbitals to form new hybrid orbitals that are more suitable for bonding.

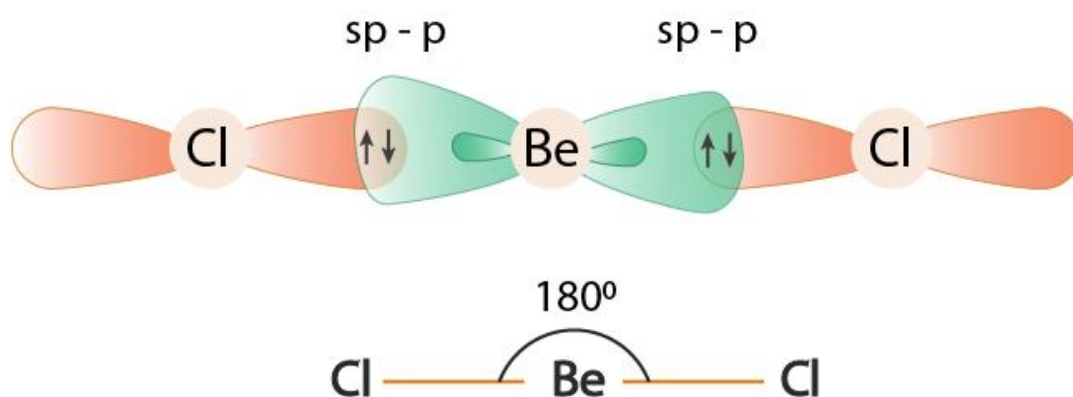
**Molecular Geometry** is the three-dimensional arrangement of atoms in a molecule. It is influenced by the number of bonding pairs and lone pairs of electrons around the central atom, as predicted by VSEPR theory.

**$\text{BeCl}_2$**

- **Hybridization:**  $sp$
- **Bond Angle:**  $180^\circ$
- **Molecular Geometry:** Linear

In  $\text{BeCl}_2$ , the beryllium atom is surrounded by two chlorine atoms. To form bonds with the chlorine atoms, the beryllium atom

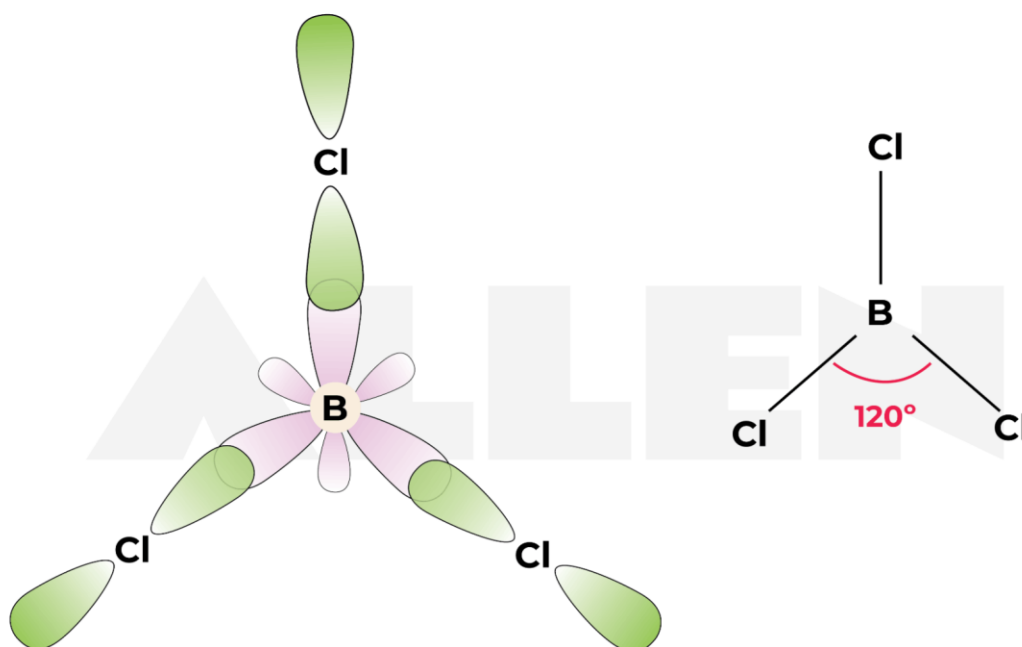
undergoes  $sp$  hybridization, mixing its  $2s$  and  $2p$  orbitals to form two equivalent  $sp$  hybrid orbitals. These hybrid orbitals are arranged linearly at  $180^\circ$  angles, resulting in a linear molecular geometry.



### $\text{BCl}_3$

- **Hybridization:**  $sp^2$
- **Bond Angle:**  $120^\circ$
- **Molecular Geometry:** Trigonal planar

In  $\text{BCl}_3$ , the boron atom is surrounded by three chlorine atoms. To form bonds with the chlorine atoms, the boron atom undergoes  $sp^2$  hybridization, mixing its  $2s$  and two  $2p$  orbitals to form three equivalent  $sp^2$  hybrid orbitals. These hybrid orbitals are arranged in a trigonal planar arrangement at  $120^\circ$  angles.

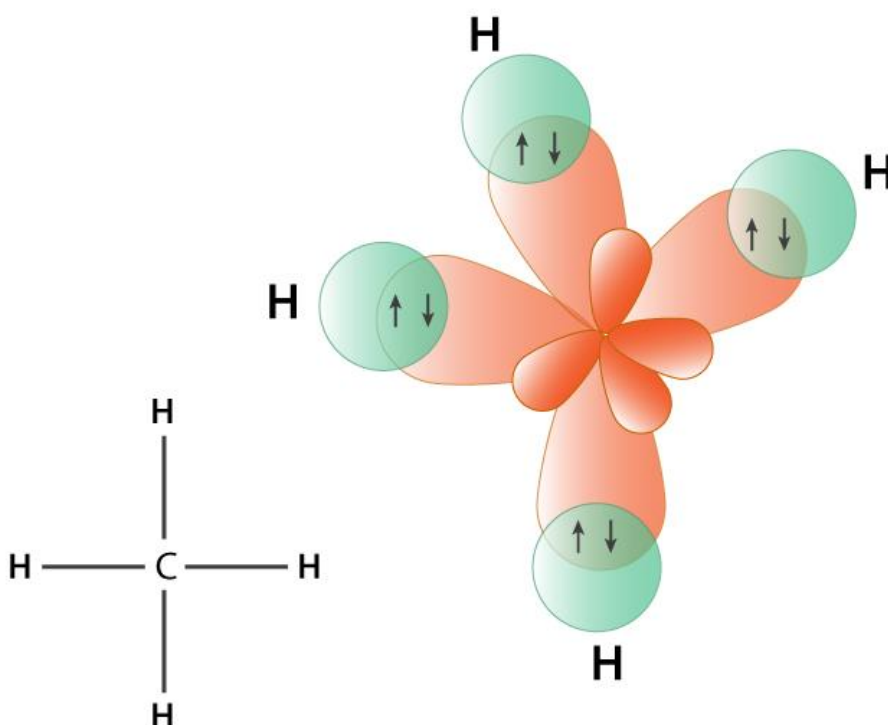


**Trigonal planar structure of  $\text{BCl}_3$**

## $\text{CH}_4$

- **Hybridization:**  $\text{sp}^3$
- **Bond Angle:**  $109.5^\circ$
- **Molecular Geometry:** Tetrahedral

In  $\text{CH}_4$ , the carbon atom is surrounded by four hydrogen atoms. To form bonds with the hydrogen atoms, the carbon atom undergoes  $\text{sp}^3$  hybridization, mixing its 2s and 2p orbitals to form four equivalent  $\text{sp}^3$  hybrid orbitals. These hybrid orbitals are arranged in a tetrahedral arrangement at  $109.5^\circ$  angles.



### Summary:

- **BeCl<sub>2</sub>:** sp hybridization, linear geometry
- **BCl<sub>3</sub>:** sp<sup>2</sup> hybridization trigonal planar geometry.
- **CH<sub>4</sub>:** sp<sup>3</sup> hybridization, tetrahedral geometry

The hybridization state of the central atom determines the molecular geometry and bond angles.

☐ **Discuss hybridisation of NH<sub>3</sub>/ H<sub>2</sub>O with it's irregular Structure and bond anigle.**

### Answer:

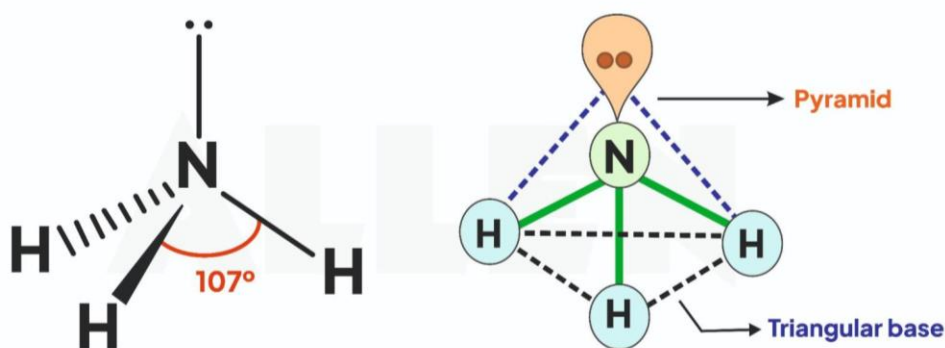
**Hybridization** is a quantum mechanical concept that explains the bonding of atoms in molecules. It involves the mixing of atomic

orbitals to form new hybrid orbitals that are more suitable for bonding.

## **NH<sub>3</sub>**

- **Hybridization:**  $sp^3$
- **Bond Angle:** Approximately  $107.3^\circ$
- **Molecular Geometry:** Trigonal pyramidal

In NH<sub>3</sub>, the nitrogen atom is surrounded by three hydrogen atoms and one lone pair of electrons. To accommodate the four electron pairs, the nitrogen atom undergoes  $sp^3$  hybridization, mixing its 2s and three 2p orbitals to form four equivalent  $sp^3$  hybrid orbitals. These hybrid orbitals are arranged in a tetrahedral arrangement. However, due to the lone pair-bonding pair repulsion, the bond angles are slightly compressed from the ideal  $109.5^\circ$  to approximately  $107.3^\circ$ .



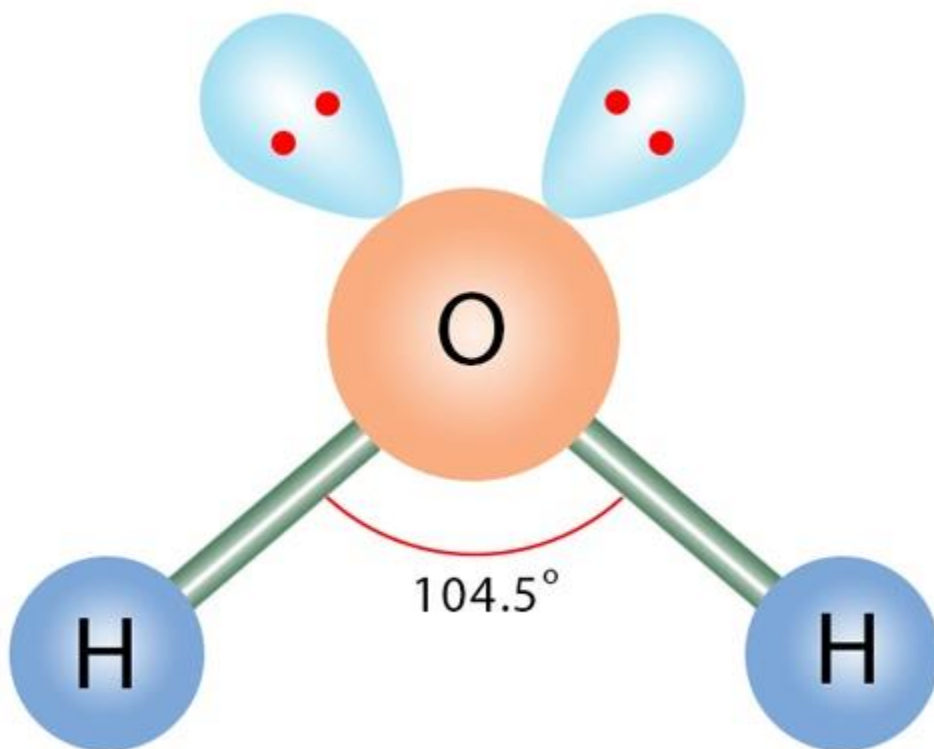
### **Bond Angle and Geometry of NH<sub>3</sub> Molecule**

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## H<sub>2</sub>O

- **Hybridization:**  $sp^3$
- **Bond Angle:** Approximately  $104.5^\circ$
- **Molecular Geometry:** Bent or angular

In H<sub>2</sub>O, the oxygen atom is surrounded by two hydrogen atoms and two lone pairs of electrons. Similar to NH<sub>3</sub>, the oxygen atom undergoes  $sp^3$  hybridization to accommodate the four electron pairs. However, the repulsion between the two lone pairs is greater than the repulsion between bonding pairs, causing the bond angle to be slightly compressed from  $109.5^\circ$  to approximately  $104.5^\circ$ .



**Summary:**

Both  $\text{NH}_3$  and  $\text{H}_2\text{O}$  exhibit  $\text{sp}^3$  hybridization, resulting in a tetrahedral arrangement of electron pairs around the central atom. However, the presence of lone pairs in these molecules causes the bond angles to deviate from the ideal  $109.5^\circ$  angle.

ありがとうございます

**(Thank you very much)**



